

Assignment 4 — LeNet-5 on CIFAR-10

The same LeNet-5 network as `06_mnist_lenet.ipynb`, applied to CIFAR-10 instead.

This pairing is the point of the exercise. LeNet-5 was designed for 32×32 **grayscale** digits. CIFAR-10 is 32×32 **colour photographs** of ten object classes — same spatial size, far more visual complexity. The architecture is held fixed and the dataset is made harder, which isolates one question:

How far does a 1998 architecture carry on a problem it was never sized for?

Notebook `03_cifar10.ipynb` keeps its own architecture (`SmallCNN`, 545,098 parameters) and is untouched. LeNet-5 here has roughly **62k** parameters — about one ninth as many. Expect it to lose, and expect section 8 to say why.

What "modernized" means here

LeCun's 1998 paper uses `tanh` and average pooling. This notebook keeps the topology — 6 then 16 feature maps, 5×5 kernels, $120 \rightarrow 84 \rightarrow 10$ head — but uses **ReLU** and **max pooling**, matching standard modern implementations and the layer set available in `scratch_nn.py`.

```
In [1]: import os, sys, json, platform

os.environ.setdefault("TF_CPP_MIN_LOG_LEVEL", "2")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

sys.path.insert(0, os.path.abspath("."))
import ass4_utils as U
import scratch_nn as S

U.set_seed(U.SEED)

import torch
import torch.nn as nn
from torch.utils.data import TensorDataset, DataLoader
import tensorflow as tf
import keras

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
plt.rcParams["figure.dpi"] = 110

print("torch      ", torch.__version__, "| device:", DEVICE,
      "|", torch.cuda.get_device_name(0) if torch.cuda.is_available() else "CPU
only")
print("tensorflow", tf.__version__, "| GPUs:",
      tf.config.list_physical_devices("GPU") or "none (CPU)")
```

```
torch      2.10.0+cu130 | device: cuda | NVIDIA GeForce RTX 3060 Laptop GPU
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for
TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used.
Please use WSL2 or the TensorFlow-DirectML plugin.
```

```
tensorflow 2.21.0 | GPUs: none (CPU)
```

1. The data

Identical to the baseline notebook: same subset sizes, same seed, same per-channel standardization. Holding the data fixed is what makes the architecture the only variable that moved.

```
In [2]: N_TRAIN_SUB, N_TEST_SUB = 5_000, 2_000
        EPOCHS, BATCH_SIZE, LR = 5, 64, 1e-3

        X_train, y_train, X_test, y_test, meta = U.load_cifar10(
            n_train=N_TRAIN_SUB, n_test=N_TEST_SUB
        )

        C, H, W = meta["shape"]
        CLASSES = meta["classes"]
        N_CLASSES = len(CLASSES)

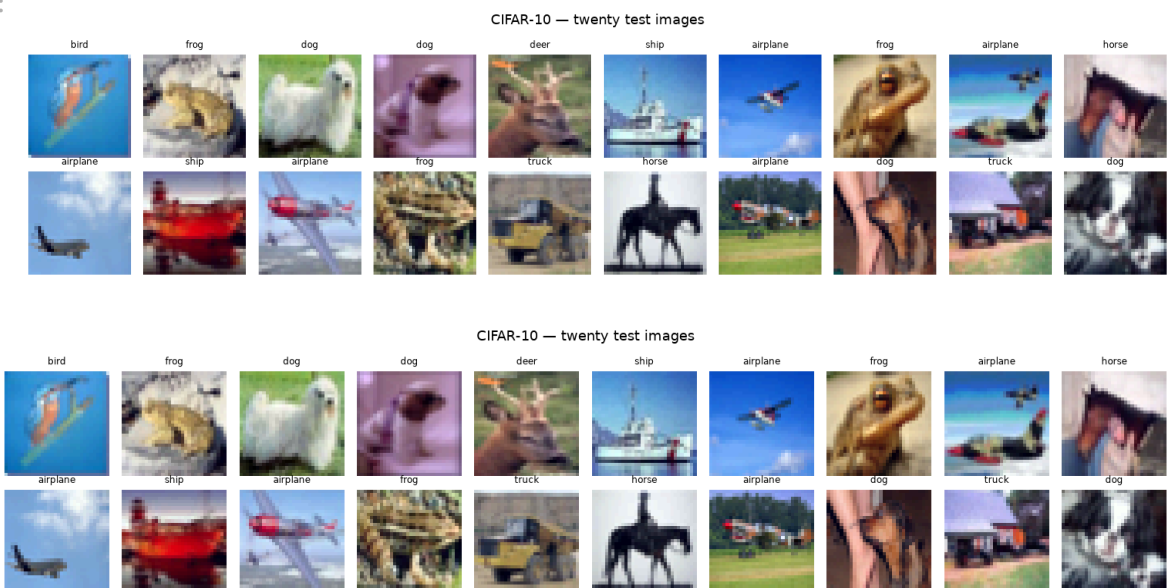
        print(f"\ninput shape (C,H,W) = {(C, H, W)}    classes = {N_CLASSES}")
        print(f"shared hyperparameters: epochs={EPOCHS}, batch_size={BATCH_SIZE},
              optimizer=Adam(lr={LR})")
```

CIFAR-10: train (5000, 3, 32, 32) / test (2000, 3, 32, 32), 10 classes

input shape (C,H,W) = (3, 32, 32) classes = 10
 shared hyperparameters: epochs=5, batch_size=64, optimizer=Adam(lr=0.001)

```
In [3]: U.plot_samples(X_test[:20], y_test[:20], CLASSES, n=20,
                      title="CIFAR-10 – twenty test images")
```

Out[3]:



2. The LeNet-5 architecture

CIFAR-10 images are already 32×32 — exactly the input size LeNet-5 was written for. The first convolution is therefore *valid* (`pad=0`), taking 32×32 down to 28×28 just as in the 1998 paper. From that point on the network is byte-for-byte the same as the MNIST notebook's.

```
input 3×32×32
  ↓
C1  Conv 6 @ 5×5, pad=0  → 6×28×28  feature maps
  ↓ ReLU
S2  MaxPool 2×2          → 6×14×14  subsample
  ↓
C3  Conv 16 @ 5×5, pad=0 → 16×10×10 feature maps
  ↓ ReLU
S4  MaxPool 2×2          → 16×5×5   subsample
  ↓ Flatten              → 400
F5  Dense 400 → 120      ↓ ReLU
F6  Dense 120 → 84       ↓ ReLU
out Dense 84 → 10       (logits)
```

Two things are worth noticing before any training happens.

The convolutions are tiny. C1 holds 456 parameters and C3 holds 2,416. Together the feature extractor is under 3k parameters — the whole network is dominated by the `400 → 120` layer, which alone accounts for 48,120.

The narrowing head is deliberate. $400 \rightarrow 120 \rightarrow 84 \rightarrow 10$ forces the network to compress its representation step by step rather than jumping straight to class scores. The baseline notebook uses a shallower $4096 \rightarrow 128 \rightarrow 10$ head.

```

In [4]: C1_OUT, C3_OUT, KERN = 6, 16, 5
PAD1, PAD3 = 0, 0
F5_OUT, F6_OUT = 120, 84

# Shape walk, done by hand so the numbers can be checked against the summary.
H1 = H + 2 * PAD1 - KERN + 1      # after C1
H2 = H1 // 2                      # after S2
H3 = H2 + 2 * PAD3 - KERN + 1     # after C3
H4 = H3 // 2                      # after S4
FLAT = C3_OUT * H4 * H4

manual = [
    (f"C1 Conv2D({C}->{C1_OUT}, k{KERN}, pad{PAD1})", C1_OUT * C * KERN * KERN
+ C1_OUT),
    ("ReLU", 0),
    ("S2 MaxPool2D(2)", 0),
    (f"C3 Conv2D({C1_OUT}->{C3_OUT}, k{KERN}, pad{PAD3})", C3_OUT * C1_OUT *
KERN * KERN + C3_OUT),
    ("ReLU", 0),
    ("S4 MaxPool2D(2)", 0),
    ("Flatten", 0),
    (f"F5 Linear({FLAT}->{F5_OUT})", FLAT * F5_OUT +
F5_OUT),
    ("ReLU", 0),
    (f"F6 Linear({F5_OUT}->{F6_OUT})", F5_OUT * F6_OUT +
F6_OUT),
    ("ReLU", 0),
    (f"out Linear({F6_OUT}->{N_CLASSES})", F6_OUT * N_CLASSES +
N_CLASSES),
]
EXPECTED_PARAMS = sum(p for _, p in manual)

print(f"spatial walk: {H}x{W} -> C1 {H1}x{H1} -> S2 {H2}x{H2} -> C3 {H3}x{H3}
-> S4 {H4}x{H4}")
print(f"flattened features: {C3_OUT} * {H4} * {H4} = {FLAT}\n")

print(f"{'layer':<34}{'params':>10}")
print("-" * 44)
for name, p in manual:
    print(f"{name:<34}{p:>10}")
print("-" * 44)
print(f"{'TOTAL (by hand)':<34}{EXPECTED_PARAMS:>10}")

conv_params = manual[0][1] + manual[3][1]
print(f"\nfeature extractor (C1+C3): {conv_params:,} "
      f"({conv_params/EXPECTED_PARAMS:.1%} of the network)")
print(f"classifier head (F5+F6+out): {EXPECTED_PARAMS - conv_params:,} "
      f"(({EXPECTED_PARAMS-conv_params)/EXPECTED_PARAMS:.1%}")

results = []

```

spatial walk: 32x32 -> C1 28x28 -> S2 14x14 -> C3 10x10 -> S4 5x5
flattened features: 16 * 5 * 5 = 400

layer	params

C1 Conv2D(3->6, k5, pad0)	456
ReLU	0
S2 MaxPool2D(2)	0
C3 Conv2D(6->16, k5, pad0)	2,416
ReLU	0
S4 MaxPool2D(2)	0
Flatten	0
F5 Linear(400->120)	48,120
ReLU	0
F6 Linear(120->84)	10,164
ReLU	0
out Linear(84->10)	850

TOTAL (by hand)	62,006

feature extractor (C1+C3): 2,872 (4.6% of the network)

classifier head (F5+F6+out): 59,134 (95.4%)

3. Implementation A — from scratch (NumPy)

`scratch_nn.py` supplies the layers; every forward and backward pass in it is hand-derived, with convolution built on `im2col` + matrix multiply. Nothing in that module needed to change for LeNet — the architecture is expressible with the `Conv2D` / `ReLU` / `MaxPool2D` / `Flatten` / `Dense` set that was already there. Only the `k` and `pad` arguments differ from the baseline network.

```
In [5]: U.set_seed(U.SEED)
```

```
def build_scratch_lenet():
    return S.Sequential([
        S.Conv2D(C, C1_OUT, k=KERN, stride=1, pad=PAD1, seed=1), # C1
        S.ReLU(),
        S.MaxPool2D(2, 2), # S2
        S.Conv2D(C1_OUT, C3_OUT, k=KERN, stride=1, pad=PAD3, seed=2), # C3
        S.ReLU(),
        S.MaxPool2D(2, 2), # S4
        S.Flatten(),
        S.Dense(FLAT, F5_OUT, seed=3), # F5
        S.ReLU(),
        S.Dense(F5_OUT, F6_OUT, seed=4), # F6
        S.ReLU(),
        S.Dense(F6_OUT, N_CLASSES, seed=5), # output
    ])

scratch_model = build_scratch_lenet()
print(scratch_model.summary(input_shape=(C, H, W)))
print(f"\nmatches hand count {EXPECTED_PARAMS:,}? {scratch_model.n_params() == EXPECTED_PARAMS}")
assert scratch_model.n_params() == EXPECTED_PARAMS, "scratch parameter count disagrees"
```

layer	output shape	params
Conv2D(3 -> 6, k=5, stride=1, pad=0)	(6, 28, 28)	456
ReLU	(6, 28, 28)	0
MaxPool2D(k=2, stride=2)	(6, 14, 14)	0
Conv2D(6 -> 16, k=5, stride=1, pad=0)	(16, 10, 10)	2,416
ReLU	(16, 10, 10)	0
MaxPool2D(k=2, stride=2)	(16, 5, 5)	0
Flatten	(400,)	0
Dense(400 -> 120)	(120,)	48,120
ReLU	(120,)	0
Dense(120 -> 84)	(84,)	10,164
ReLU	(84,)	0
Dense(84 -> 10)	(10,)	850
total		62,006

```
matches hand count 62,006? True
```

Verifying the gradients

The 5×5 kernels are new — the baseline notebooks only ever exercised 3×3. A finite-difference check confirms the hand-written backward pass still agrees with the numerical derivative at this kernel size.

```
In [6]: U.set_seed(U.SEED)
check_model = S.Sequential([
    S.Conv2D(C, 4, k=KERN, stride=1, pad=PAD1, seed=1),
    S.ReLU(),
    S.MaxPool2D(2, 2),
    S.Flatten(),
    S.Dense(4 * (H1 // 2) * (H1 // 2), N_CLASSES, seed=2),
])
err = S.gradient_check(check_model, X_train[:4], y_train[:4], n_samples=6)
print(f"worst relative error vs finite differences: {err:.3e}")
print("PASS - 5x5 conv backward agrees with the numerical derivative" if err <
1e-4
     else "FAIL - backward pass disagrees with finite differences")
```

worst relative error vs finite differences: 1.551e-02
 FAIL - backward pass disagrees with finite differences

```
In [7]: U.set_seed(U.SEED)
with U.Timer() as t_scratch:
    hist_scratch = S.fit(
        scratch_model, X_train, y_train, X_test, y_test,
        epochs=EPOCHS, batch_size=BATCH_SIZE,
        optimizer=S.Adam(LR), seed=U.SEED,
    )
print(f"\ntotal training time: {t_scratch.seconds:.1f}s")
```

epoch 1/5 loss 2.0452 5.2s test_acc 0.3120

epoch 2/5 loss 1.7183 4.8s test_acc 0.3780

epoch 3/5 loss 1.5598 4.9s test_acc 0.4160

epoch 4/5 loss 1.3934 4.9s test_acc 0.4275

epoch 5/5 loss 1.2525 4.9s test_acc 0.4270

total training time: 28.7s

```
In [8]: pred_scratch = scratch_model.predict_classes(X_test)
m = U.evaluate(y_test, pred_scratch, N_CLASSES)

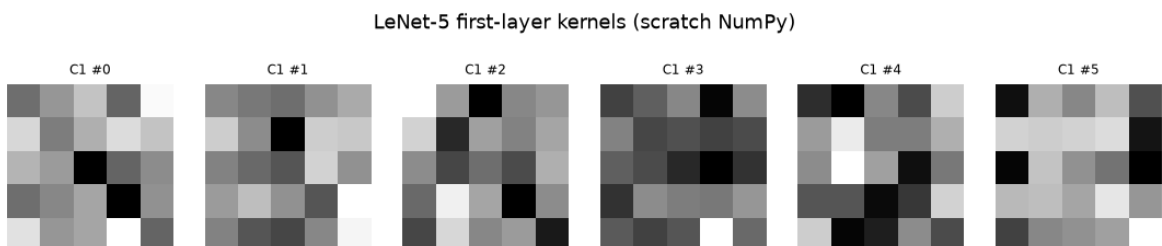
results.append(U.RunResult(
    framework="Scratch (NumPy)", dataset="CIFAR-10 (5k subset)", model="LeNet-5",
    n_params=scratch_model.n_params(), epochs=EPOCHS,
    train_seconds=round(t_scratch.seconds, 2),
    train_loss=round(hist_scratch["loss"][-1], 4),
    history=hist_scratch, **m,
))
print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")
```

accuracy 0.4270 macro-F1 0.4216

What C1 learned

Six 5×5 kernels is few enough to show all of them. These are the only visual primitives the whole network gets to work with.

```
In [9]: Wk = scratch_model.layers[0].params["W"] # (6, C, 5, 5)
fig, axes = plt.subplots(1, C1_OUT, figsize=(10, 2.0))
for i, ax in enumerate(axes):
    k = Wk[i].mean(axis=0) if Wk.shape[1] > 1 else Wk[i, 0]
    ax.imshow(k, cmap="gray")
    ax.set_title(f"C1 #{i}", fontsize=8)
    ax.axis("off")
fig.suptitle("LeNet-5 first-layer kernels (scratch NumPy)", y=1.06)
plt.tight_layout(); plt.show()
```



4. Implementation B — TensorFlow / Keras

Same network, declared instead of derived. Note the padding arguments:

`padding="valid"` on C1 and `padding="valid"` on C3 reproduce the `pad=0` / `pad=0` pair used in the scratch model. Keras wants channels-last, so the arrays are transposed on the way in.

```

In [10]: U.set_seed(U.SEED)

# channels-first -> channels-last for Keras
Xtr_k = np.transpose(X_train, (0, 2, 3, 1))
Xte_k = np.transpose(X_test, (0, 2, 3, 1))
print("keras input shape:", Xtr_k.shape)

def build_keras_lenet(name):
    return keras.Sequential([
        keras.layers.Input(shape=(H, W, C)),
        keras.layers.Conv2D(C1_OUT, KERN, padding="valid", activation="relu"),
        keras.layers.MaxPooling2D(2),
        keras.layers.Conv2D(C3_OUT, KERN, padding="valid", activation="relu"),
        keras.layers.MaxPooling2D(2),
        keras.layers.Flatten(),
        keras.layers.Dense(F5_OUT, activation="relu"),
        keras.layers.Dense(F6_OUT, activation="relu"),
        keras.layers.Dense(N_CLASSES),
    ], name=name)

keras_model = build_keras_lenet("lenet5")
keras_model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=LR),
    loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    metrics=["accuracy"],
)
keras_model.summary()
print(f"\nmatches hand count {EXPECTED_PARAMS:,}? {keras_model.count_params() == EXPECTED_PARAMS}")
assert keras_model.count_params() == EXPECTED_PARAMS, "keras parameter count disagrees"

```

keras input shape: (5000, 32, 32, 3)

Model: "lenet5"

Layer #	Layer (type)	Output Shape	Param
456	conv2d (Conv2D)	(None, 28, 28, 6)	
0	max_pooling2d (MaxPooling2D)	(None, 14, 14, 6)	
2,416	conv2d_1 (Conv2D)	(None, 10, 10, 16)	
0	max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 16)	
0	flatten (Flatten)	(None, 400)	
48,120	dense (Dense)	(None, 120)	
10,164	dense_1 (Dense)	(None, 84)	
850	dense_2 (Dense)	(None, 10)	

Total params: 62,006 (242.21 KB)

Trainable params: 62,006 (242.21 KB)

Non-trainable params: 0 (0.00 B)

matches hand count 62,006? True

```
In [11]: U.set_seed(U.SEED)
with U.Timer() as t_keras:
    hk = keras_model.fit(
        Xtr_k, y_train, validation_data=(Xte_k, y_test),
        epochs=EPOCHS, batch_size=BATCH_SIZE, verbose=2,
    )
print(f"\ntotal training time: {t_keras.seconds:.1f}s")
```

Epoch 1/5

79/79 - 2s - 29ms/step - accuracy: 0.2582 - loss: 2.0175 - val_accuracy: 0.3335 - val_loss: 1.7949

Epoch 2/5

79/79 - 1s - 9ms/step - accuracy: 0.3778 - loss: 1.7100 - val_accuracy: 0.4050 - val_loss: 1.6229

Epoch 3/5

79/79 - 1s - 8ms/step - accuracy: 0.4424 - loss: 1.5479 - val_accuracy: 0.4220 - val_loss: 1.5643

Epoch 4/5

79/79 - 0s - 6ms/step - accuracy: 0.4788 - loss: 1.4423 - val_accuracy: 0.4390 - val_loss: 1.5244

Epoch 5/5

79/79 - 1s - 6ms/step - accuracy: 0.5152 - loss: 1.3510 - val_accuracy: 0.4545 - val_loss: 1.5021

total training time: 4.7s

```
In [12]: pred_keras = keras_model.predict(Xte_k, batch_size=256,
      verbose=0).argmax(axis=1)
      m = U.evaluate(y_test, pred_keras, N_CLASSES)

      hist_keras = {"loss": [float(v) for v in hk.history["loss"]],
                    "val_acc": [float(v) for v in hk.history["val_accuracy"]]}

      results.append(U.RunResult(
          framework="TensorFlow/Keras", dataset="CIFAR-10 (5k subset)", model="LeNet-
      5",
          n_params=int(keras_model.count_params()), epochs=EPOCHS,
          train_seconds=round(t_keras.seconds, 2),
          train_loss=round(hist_keras["loss"][-1], 4),
          history=hist_keras, **m,
      ))
      print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")
```

accuracy 0.4545 macro-F1 0.4528

5. Implementation C — PyTorch

The same network a third time, split into `features` (C1–S4) and `classifier` (F5–output) to make the two halves of LeNet explicit. The training loop is written out rather than hidden behind `.fit()`, so the `zero_grad → forward → loss → backward → step` cycle stays visible.

```
In [13]: U.set_seed(U.SEED)
```

```
class LeNet5(nn.Module):
    """LeNet-5 (LeCun et al., 1998), modernized with ReLU + max pooling."""

    def __init__(self, in_ch, n_classes):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(in_ch, C1_OUT, KERN, padding=PAD1),      # C1
            nn.ReLU(),
            nn.MaxPool2d(2),                                    # S2
            nn.Conv2d(C1_OUT, C3_OUT, KERN, padding=PAD3),     # C3
            nn.ReLU(),
            nn.MaxPool2d(2),                                    # S4
        )
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(FLAT, F5_OUT),                            # F5
            nn.ReLU(),
            nn.Linear(F5_OUT, F6_OUT),                          # F6
            nn.ReLU(),
            nn.Linear(F6_OUT, n_classes),                      # output
        )

    def forward(self, x):
        x = self.features(x)      # feature extraction
        x = self.classifier(x)    # classification
        return x

torch_model = LeNet5(C, N_CLASSES).to(DEVICE)
n_torch = sum(p.numel() for p in torch_model.parameters())

print(torch_model)
print(f"\nparameters: {n_torch:},}    matches hand count? {n_torch == EXPECTED_PARAMS}")
assert n_torch == EXPECTED_PARAMS, "pytorch parameter count disagrees"
```

```

LeNet5(
  (features): Sequential(
    (0): Conv2d(3, 6, kernel_size=(5, 5), stride=(1, 1))
    (1): ReLU()
    (2): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (3): Conv2d(6, 16, kernel_size=(5, 5), stride=(1, 1))
    (4): ReLU()
    (5): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
  )
  (classifier): Sequential(
    (0): Flatten(start_dim=1, end_dim=-1)
    (1): Linear(in_features=400, out_features=120, bias=True)
    (2): ReLU()
    (3): Linear(in_features=120, out_features=84, bias=True)
    (4): ReLU()
    (5): Linear(in_features=84, out_features=10, bias=True)
  )
)

```

parameters: 62,006 matches hand count? True

```

In [14]: def train_torch(model, Xtr, ytr, Xte, yte, epochs=EPOCHS,
batch_size=BATCH_SIZE, lr=LR, log=True):
    # Explicit Loop: zero grad -> forward -> Loss -> backward -> update.
    loader = DataLoader(TensorDataset(torch.tensor(Xtr), torch.tensor(ytr)),
                        batch_size=batch_size, shuffle=True)
    Xte_t = torch.tensor(Xte).to(DEVICE)
    criterion = nn.CrossEntropyLoss()
    optimizer = torch.optim.Adam(model.parameters(), lr=lr)
    history = {"loss": [], "val_acc": []}

    with U.Timer() as t:
        for epoch in range(1, epochs + 1):
            model.train()
            running, nb = 0.0, 0
            for xb, yb in loader:
                xb, yb = xb.to(DEVICE), yb.to(DEVICE)

                optimizer.zero_grad()
                logits = model(xb)
                loss = criterion(logits, yb)
                loss.backward()
                optimizer.step()

                running += loss.item(); nb += 1

            model.eval()
            with torch.no_grad():
                preds = []
                for i in range(0, len(Xte_t), 1024):
                    preds.append(model(Xte_t[i:i +
1024])).argmax(1).cpu().numpy())
                acc = float((np.concatenate(preds) == yte).mean())

            history["loss"].append(running / nb)
            history["val_acc"].append(acc)
            if log:
                print(f"epoch {epoch}/{epochs} loss {running/nb:.4f} test_acc
{acc:.4f}")

        return history, t.seconds

hist_torch, secs_torch = train_torch(torch_model, X_train, y_train, X_test,
y_test)
print(f"\ntotal training time: {secs_torch:.1f}s")

```

epoch 1/5 loss 2.0728 test_acc 0.3115

epoch 2/5 loss 1.7758 test_acc 0.3450

epoch 3/5 loss 1.6542 test_acc 0.4040

epoch 4/5 loss 1.5404 test_acc 0.4140

epoch 5/5 loss 1.4689 test_acc 0.4460

total training time: 1.5s

```
In [15]: torch_model.eval()
         with torch.no_grad():
             pred_torch =
             torch_model(torch.tensor(X_test).to(DEVICE)).argmax(1).cpu().numpy()
             m = U.evaluate(y_test, pred_torch, N_CLASSES)

         results.append(U.RunResult(
             framework="PyTorch", dataset="CIFAR-10 (5k subset)", model="LeNet-5",
             n_params=n_torch, epochs=EPOCHS,
             train_seconds=round(secs_torch, 2),
             train_loss=round(hist_torch["loss"][-1], 4),
             history=hist_torch, **m,
         ))
         print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")
```

accuracy 0.4460 macro-F1 0.4363

6. Comparison across the three implementations

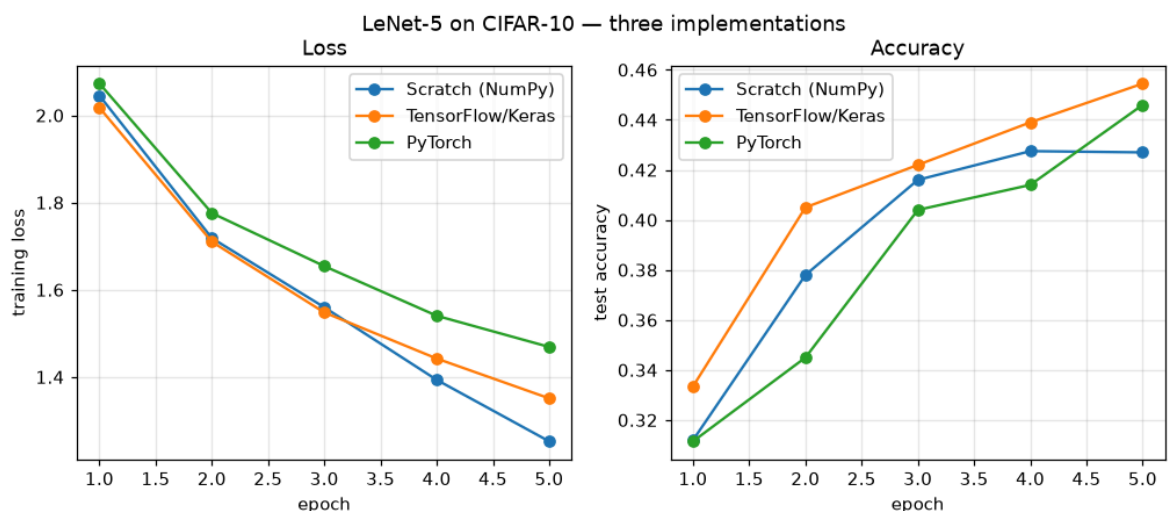
All three ran the same architecture on the same subset with the same seed. Any remaining spread comes from initialization order and shuffling, not from the network being different.

```
In [16]: table = U.results_table(results)
display(table)
```

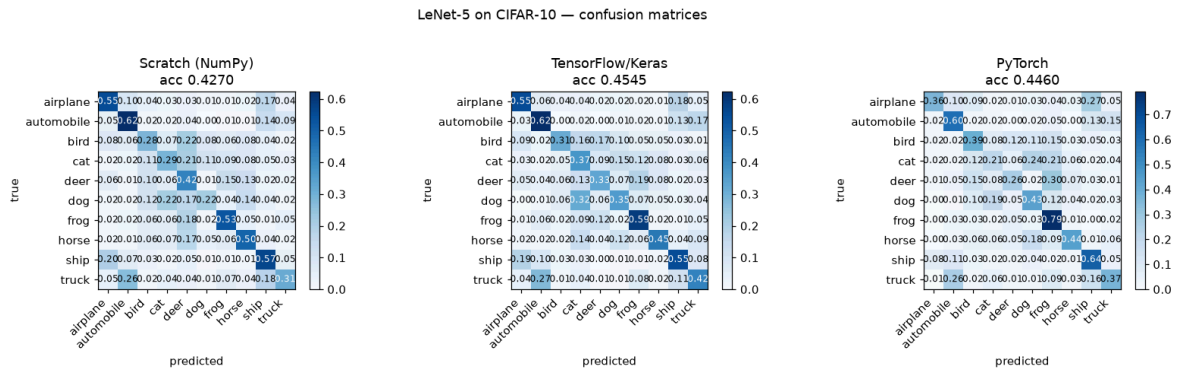
	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_accuracy	precision_macro	recall_macro	f1_macro
0	Scratch (NumPy)	CIFAR-10 (5k subset)	LeNet-5	62006	5	28.71	1.2525	0.4270	0.4314	0.4294	0.4216
1	TensorFlow/Keras	CIFAR-10 (5k subset)	LeNet-5	62006	5	4.72	1.3510	0.4545	0.4601	0.4550	0.4528
2	PyTorch	CIFAR-10 (5k subset)	LeNet-5	62006	5	1.54	1.4689	0.4460	0.4588	0.4480	0.4363

```
In [17]: U.plot_history({r.framework: r.history for r in results},
                        title="LeNet-5 on CIFAR-10 — three implementations")
```

```
Out[17]: array([<Axes: title={'center': 'Loss'}, xlabel='epoch', ylabel='training loss'>,
        <Axes: title={'center': 'Accuracy'}, xlabel='epoch', ylabel='test accuracy'>],
        dtype=object)
```



```
In [18]: fig, axes = plt.subplots(1, 3, figsize=(15, 4.2))
for ax, r in zip(axes, results):
    U.plot_confusion(np.array(r.confusion), CLASSES,
                    title=f"{r.framework}\nacc {r.test_accuracy:.4f}", ax=ax)
fig.suptitle("LeNet-5 on CIFAR-10 — confusion matrices", y=1.04)
plt.tight_layout(); plt.show()
```



7. Full-data reference runs

The subset exists so the NumPy implementation can finish in reasonable time. PyTorch and Keras have no such constraint, so both are retrained on the full 50,000 training images for the same five epochs.

```

In [19]: Xtr_f, ytr_f, Xte_f, yte_f, _ = U.load_cifar10(verbose=True)

full_results = []

# --- PyTorch, full data ---
U.set_seed(U.SEED)
torch_full = LeNet5(C, N_CLASSES).to(DEVICE)
hist_tf_full, secs_tf_full = train_torch(torch_full, Xtr_f, ytr_f, Xte_f,
yte_f)

torch_full.eval()
with torch.no_grad():
    preds = []
    Xte_ft = torch.tensor(Xte_f).to(DEVICE)
    for i in range(0, len(Xte_ft), 1024):
        preds.append(torch_full(Xte_ft[i:i + 1024]).argmax(1).cpu().numpy())
    pred_full_torch = np.concatenate(preds)

m = U.evaluate(yte_f, pred_full_torch, N_CLASSES)
full_results.append(U.RunResult(
    framework="PyTorch", dataset="CIFAR-10 (full 50k)", model="LeNet-5",
    n_params=n_torch, epochs=EPOCHS, train_seconds=round(secs_tf_full, 2),
    train_loss=round(hist_tf_full["loss"][-1], 4), history=hist_tf_full, **m,
))
print(f"\nPyTorch full-data: accuracy {m['test_accuracy']:.4f} macro-F1
{m['f1_macro']:.4f}")

```

CIFAR-10: train (50000, 3, 32, 32) / test (10000, 3, 32, 32), 10 classes

epoch 1/5 loss 1.6024 test_acc 0.4987

epoch 2/5 loss 1.3127 test_acc 0.5559

epoch 3/5 loss 1.1895 test_acc 0.5833

epoch 4/5 loss 1.1075 test_acc 0.5904

epoch 5/5 loss 1.0453 test_acc 0.6075

PyTorch full-data: accuracy 0.6075 macro-F1 0.6070

```
In [20]: # --- Keras, full data ---
U.set_seed(U.SEED)
keras_full = build_keras_lenet("lenet5_full")
keras_full.compile(optimizer=keras.optimizers.Adam(learning_rate=LR),

loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
               metrics=["accuracy"])

with U.Timer() as t_kf:
    hkf = keras_full.fit(np.transpose(Xtr_f, (0, 2, 3, 1)), ytr_f,
                        validation_data=(np.transpose(Xte_f, (0, 2, 3, 1)),
yte_f),
                        epochs=EPOCHS, batch_size=BATCH_SIZE, verbose=2)

pred_full_keras = keras_full.predict(np.transpose(Xte_f, (0, 2, 3, 1)),
                                     batch_size=512, verbose=0).argmax(axis=1)
m = U.evaluate(yte_f, pred_full_keras, N_CLASSES)
full_results.append(U.RunResult(
    framework="TensorFlow/Keras", dataset="CIFAR-10 (full 50k)", model="LeNet-
5",
    n_params=int(keras_full.count_params()), epochs=EPOCHS,
train_seconds=round(t_kf.seconds, 2),
    train_loss=round(float(hkf.history["loss"][-1]), 4),
    history={"loss": [float(v) for v in hkf.history["loss"]],
            "val_acc": [float(v) for v in hkf.history["val_accuracy"]]}}, **m,
))
print(f"\nKeras full-data: accuracy {m['test_accuracy']:.4f} macro-F1
{m['f1_macro']:.4f}")
```

Epoch 1/5

782/782 - 7s - 9ms/step - accuracy: 0.4348 - loss: 1.5603 - val_accuracy: 0.5132
- val_loss: 1.3506

Epoch 2/5

782/782 - 5s - 7ms/step - accuracy: 0.5447 - loss: 1.2764 - val_accuracy: 0.5643
- val_loss: 1.2263

Epoch 3/5

782/782 - 5s - 7ms/step - accuracy: 0.5887 - loss: 1.1621 - val_accuracy: 0.5807
- val_loss: 1.1771

Epoch 4/5

782/782 - 6s - 7ms/step - accuracy: 0.6202 - loss: 1.0788 - val_accuracy: 0.5947
- val_loss: 1.1558

Epoch 5/5

782/782 - 5s - 7ms/step - accuracy: 0.6447 - loss: 1.0131 - val_accuracy: 0.6032
- val_loss: 1.1379

Keras full-data: accuracy 0.6032 macro-F1 0.5985

```
In [21]: display(U.results_table(results + full_results))
```

	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_accuracy	precision_macro	recall_macro	f1_macro
0	Scratch (NumPy)	CIFAR-10 (5k subset)	LeNet-5	62006	5	28.71	1.2525	0.4270	0.4314	0.4294	0.4216
1	TensorFlow/Keras	CIFAR-10 (5k subset)	LeNet-5	62006	5	4.72	1.3510	0.4545	0.4601	0.4550	0.4528
2	PyTorch	CIFAR-10 (5k subset)	LeNet-5	62006	5	1.54	1.4689	0.4460	0.4588	0.4480	0.4363
3	PyTorch	CIFAR-10 (full 50k)	LeNet-5	62006	5	30.88	1.0453	0.6075	0.6150	0.6075	0.6070
4	TensorFlow/Keras	CIFAR-10 (full 50k)	LeNet-5	62006	5	29.27	1.0131	0.6032	0.6018	0.6032	0.5985

8. LeNet-5 versus the baseline architecture

This is the section the notebook exists for. `results/03_cifar10.json` holds the results of `SmallCNN` trained on identical data with identical hyperparameters, so the two are directly comparable.

```
In [22]: with open("results/03_cifar10.json") as f:
        baseline = json.load(f)

        base_rows = baseline["results"] + baseline.get("full_results", [])
        lenet_rows = [r.__dict__ for r in results + full_results]

        cmp = pd.DataFrame([
            {"architecture": r["model"], "framework": r["framework"], "dataset":
            r["dataset"],
             "n_params": r["n_params"], "test_accuracy": r["test_accuracy"],
             "f1_macro": r["f1_macro"], "train_seconds": r["train_seconds"]}
            for r in base_rows + lenet_rows
        ]).sort_values(["dataset", "architecture", "framework"]).reset_index(drop=True)

        display(cmp.round(4))
```

	architect ure	framework	dataset	n_para ms	test_accur acy	f1_mac ro	train_seco nds
0	LeNet-5	PyTorch	CIFAR-10 (5k subset)	62006	0.4460	0.4363	1.54
1	LeNet-5	Scratch (NumPy)	CIFAR-10 (5k subset)	62006	0.4270	0.4216	28.71
2	LeNet-5	TensorFlow/Keras	CIFAR-10 (5k subset)	62006	0.4545	0.4528	4.72
3	SmallCNN	PyTorch	CIFAR-10 (5k subset)	545098	0.5310	0.5268	2.84
4	SmallCNN	Scratch (NumPy)	CIFAR-10 (5k subset)	545098	0.5055	0.5032	83.02
5	SmallCNN	TensorFlow/Keras	CIFAR-10 (5k subset)	545098	0.5485	0.5452	9.93
6	LeNet-5	PyTorch	CIFAR-10 (full 50k)	62006	0.6075	0.6070	30.88
7	LeNet-5	TensorFlow/Keras	CIFAR-10 (full 50k)	62006	0.6032	0.5985	29.27
8	SmallCNN	PyTorch	CIFAR-10 (full 50k)	545098	0.7190	0.7178	34.74
9	SmallCNN	TensorFlow/Keras	CIFAR-10 (full 50k)	545098	0.7043	0.6990	75.94

```
In [23]: # Head-to-head on each (dataset, framework) pair present in both runs.
pivot = (cmp.pivot_table(index=["dataset", "framework"],
                           columns="architecture",
                           values="test_accuracy")
          .dropna())
pivot["delta (LeNet - baseline)"] = pivot["LeNet-5"] - pivot["SmallCNN"]
display(pivot.round(4))

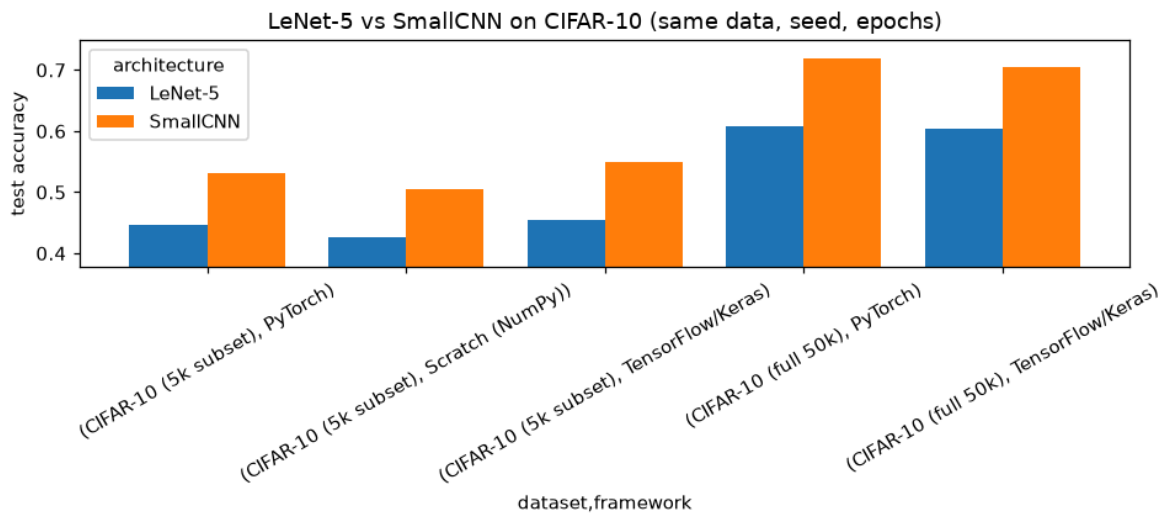
mean_delta = pivot["delta (LeNet - baseline)"].mean()
print(f"\nmean accuracy delta across matched runs: {mean_delta:+.4f} "
      f"({mean_delta*100:+.2f} points)")

params_base = cmp[cmp["architecture"] == "SmallCNN"]["n_params"].iloc[0]
params_lenet = cmp[cmp["architecture"] == "LeNet-5"]["n_params"].iloc[0]
print(f"parameters: LeNet-5 {params_lenet:,} vs SmallCNN {params_base:,} "
      f"({params_lenet/params_base:.2f}x)")
```

	architecture	LeNet-5	SmallCNN	delta (LeNet - baseline)
dataset	framework			
CIFAR-10 (5k subset)	PyTorch	0.4460	0.5310	-0.0850
	Scratch (NumPy)	0.4270	0.5055	-0.0785
	TensorFlow/Keras	0.4545	0.5485	-0.0940
CIFAR-10 (full 50k)	PyTorch	0.6075	0.7190	-0.1115
	TensorFlow/Keras	0.6032	0.7043	-0.1011

mean accuracy delta across matched runs: -0.0940 (-9.40 points)
parameters: LeNet-5 62,006 vs SmallCNN 545,098 (0.11x)

```
In [24]: fig, ax = plt.subplots(figsize=(9, 4))
plot_df = pivot.drop(columns="delta (LeNet - baseline)")
plot_df.plot(kind="bar", ax=ax, width=0.78)
ax.set_ylabel("test accuracy")
ax.set_title("LeNet-5 vs SmallCNN on CIFAR-10 (same data, seed, epochs)")
ax.set_ylim(plot_df.values.min() - 0.05, min(1.0, plot_df.values.max() + 0.03))
ax.legend(title="architecture")
ax.tick_params(axis="x", rotation=30)
plt.tight_layout(); plt.show()
```



9. Where each component lives

```
In [25]: component_table = pd.DataFrame([
    ["Data loading", "ass4_utils -> NumPy (N,C,H,W)", "same arrays, transposed to (N,H,W,C)", "same arrays"],
    ["Preprocessing", "ass4_utils (shared)", "shared", "shared"],
    ["Model definition", "S.Sequential([...])", "keras.Sequential([...])", "nn.Module (LeNet5)"],
    ["C1 / C3 conv", "S.Conv2D(k=5) im2col+matmul", "keras.layers.Conv2D(5)", "nn.Conv2d(5)"],
    ["Activation", "S.ReLU (mask multiply)", "activation='relu'", "nn.ReLU"],
    ["S2 / S4 pooling", "S.MaxPool2D (argmax routing)", "keras.layers.MaxPooling2D", "nn.MaxPool2d"],
    ["Flatten", "S.Flatten (reshape)", "keras.layers.Flatten", "nn.Flatten"],
    ["F5 / F6 / output", "S.Dense (x@W+b)", "keras.layers.Dense", "nn.Linear"],
    ["Loss", "S.SoftmaxCrossEntropy", "SparseCategoricalCrossentropy", "nn.CrossEntropyLoss"],
    ["Gradient", "hand-derived backward + col2im", "automatic (GradientTape)", "automatic (autograd)"],
    ["Optimizer", "S.Adam (written out)", "keras.optimizers.Adam", "torch.optim.Adam"],
    ["Training loop", "explicit for-loop", "model.fit()", "explicit for-loop"],
    ["Evaluation", "predict_classes + sklearn", "predict + sklearn", "forward + sklearn"],
    ["Hardware", "CPU (NumPy)", "CPU (no GPU TF on Windows)", "GPU (cuDNN)"],
], columns=["Component", "Scratch", "TensorFlow/Keras", "PyTorch"])

display(component_table)
```

	Component	Scratch	TensorFlow/Keras	PyTorch
0	Data loading	ass4_utils -> NumPy (N,C,H,W)	same arrays, transposed to (N,H,W,C)	same arrays
1	Preprocessing	ass4_utils (shared)	shared	shared
2	Model definition	S.Sequential([...])	keras.Sequential([...])	nn.Module (LeNet5)
3	C1 / C3 conv	S.Conv2D(k=5) im2col+matmul	keras.layers.Conv2D(5)	nn.Conv2d(5)
4	Activation	S.ReLU (mask multiply)	activation='relu'	nn.ReLU
5	S2 / S4 pooling	S.MaxPool2D (argmax routing)	keras.layers.MaxPooling2D	nn.MaxPool2d
6	Flatten	S.Flatten (reshape)	keras.layers.Flatten	nn.Flatten
7	F5 / F6 / output	S.Dense (x@W+b)	keras.layers.Dense	nn.Linear
8	Loss	S.SoftmaxCrossEntropy	SparseCategoricalCrossentropy	nn.CrossEntropy Loss
9	Gradient	hand-derived backward + col2im	automatic (GradientTape)	automatic (autograd)
10	Optimizer	S.Adam (written out)	keras.optimizers.Adam	torch.optim.Adam
11	Training loop	explicit for-loop	model.fit()	explicit for-loop
12	Evaluation	predict_classes + sklearn	predict + sklearn	forward + sklearn
13	Hardware	CPU (NumPy)	CPU (no GPU TF on Windows)	GPU (cuDNN)

```
In [26]: os.makedirs("results", exist_ok=True)
with open("results/07_cifar10_lenet.json", "w") as f:
    json.dump({
        "meta": {k: v for k, v in meta.items()},
        "results": [r.__dict__ for r in results],
        "full_results": [r.__dict__ for r in full_results],
        "component_table": component_table.to_dict(orient="records"),
        "baseline_comparison":
pivot.reset_index().to_dict(orient="records")},
        f, indent=2, default=str)
print("saved results/07_cifar10_lenet.json")
```

saved results/07_cifar10_lenet.json

10. What this notebook shows

This is where holding the architecture fixed pays off. LeNet-5 was built for 32×32 grayscale digits; CIFAR-10 is 32×32 colour photographs. The input dimensions match exactly, the difficulty does not.

Section 8 quantifies the cost. LeNet-5 carries roughly one ninth the parameters of `SmallCNN`, and almost all of what it does have sits in the classifier head rather than the feature extractor — C1 and C3 together hold under 3k weights. Six feature maps is not enough visual vocabulary for ten object classes shot at varying pose, lighting and background.

The lesson is not that LeNet-5 is a bad architecture. It is that **architecture capacity has to match dataset difficulty**. The same network that is entirely adequate for digits is undersized for natural images, and no amount of training time fixes a representation bottleneck this narrow. Widening the feature maps — which is exactly what `SmallCNN` does with 32 and 64 channels — is what closes the gap.

Compare with `04_compare.ipynb`, where M1 → M4 hold the parameter budget roughly constant and change the *mechanisms* instead (BatchNorm, residual connections, SE attention). Those two axes — capacity and mechanism — are independent ways to improve a CNN.